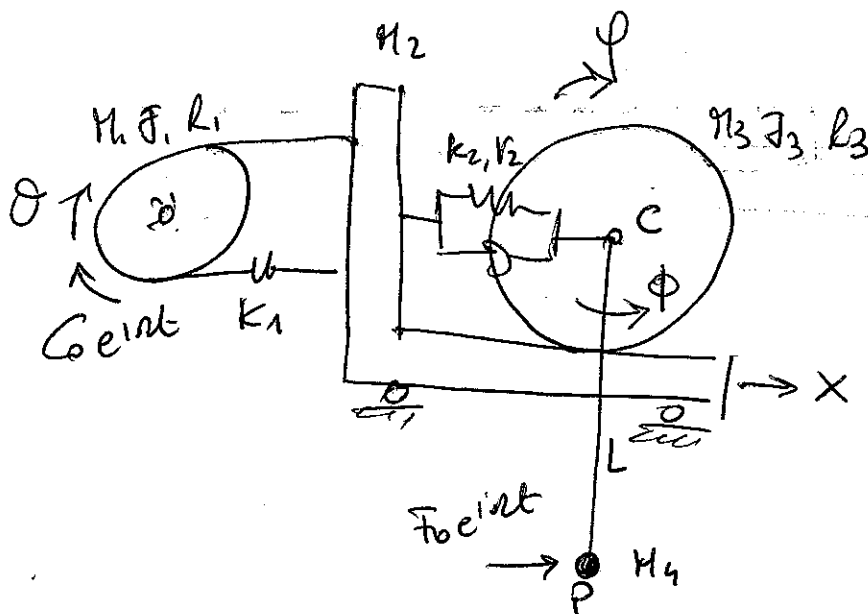


28 Jan 2008



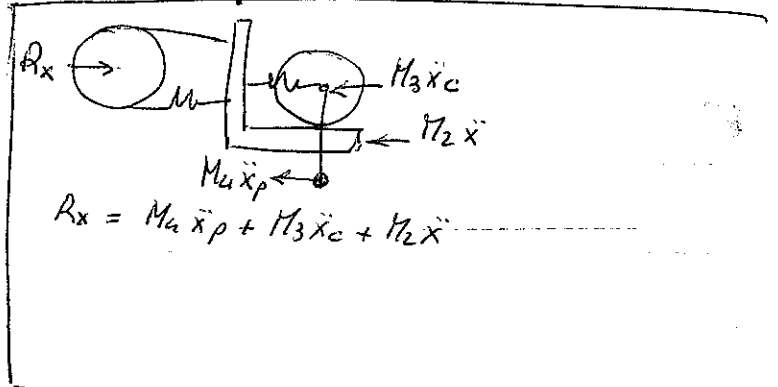
$$\underline{x} = \begin{Bmatrix} x \\ \phi \\ \phi \end{Bmatrix}$$

$$\bar{E}_c = \frac{1}{2} \bar{J}_1 \dot{\theta}^2 + \frac{1}{2} M_2 \dot{x}^2 + \frac{1}{2} M_3 v_c^2 + \frac{1}{2} \bar{J}_3 \dot{\phi}^2 + \frac{1}{2} M_4 v_p^2$$

$$[m] = \text{diag}(\bar{J}_1, M_2, M_3, \bar{J}_3, M_4) \quad \underline{y}_{WF} = \{\theta, x, x_c, \phi, x_p\}^T$$

$$\theta = x/R_1 \quad x_c = x + R_3 \phi \quad x_p = x + R_3 \phi + L \phi$$

$$[J_{WF}] = \begin{bmatrix} 1/R_1 & 0 & 0 \\ 1 & R_3 & 0 \\ 0 & 1 & 0 \\ 1 & R_3 & L \end{bmatrix}$$



$$V_{EL} = \frac{1}{2} k_1 \Delta l_1^2 + \frac{1}{2} k_2 \Delta l_2^2, \quad [k_{EL}] = \text{diag}(k_1, k_2)$$

$$\Delta l_1 = 2x \quad \Delta l_2 = R_3 \phi \quad [r_F] = r_2$$

$$[J_{EL}] = \begin{bmatrix} 2 & 0 & 0 \\ 0 & R_3 & 0 \end{bmatrix} \quad [J_r] = \begin{bmatrix} 0 & R_3 & 0 \end{bmatrix}$$

$$V_s = M_4 g (-L \cos \phi) \rightarrow \frac{\partial^2 V_s}{\partial \phi^2} \bigg|_0 = M_4 g L \rightarrow [k_s] = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & M_4 g L \end{bmatrix}$$

$$\delta L = F \delta x_p + C \delta \theta = F(\delta x + R_3 \delta \phi + L \delta \phi) + C \frac{\delta x}{R_1}$$

$$\underline{Q}_F = \begin{Bmatrix} 1 & R_3 & L \end{Bmatrix}^T F_{0cint} \quad \underline{Q}_C = \begin{Bmatrix} 1/R_1 & 0 & 0 \end{Bmatrix}^T C_{cint}$$